Self-Correction Benchmark Results, Explained

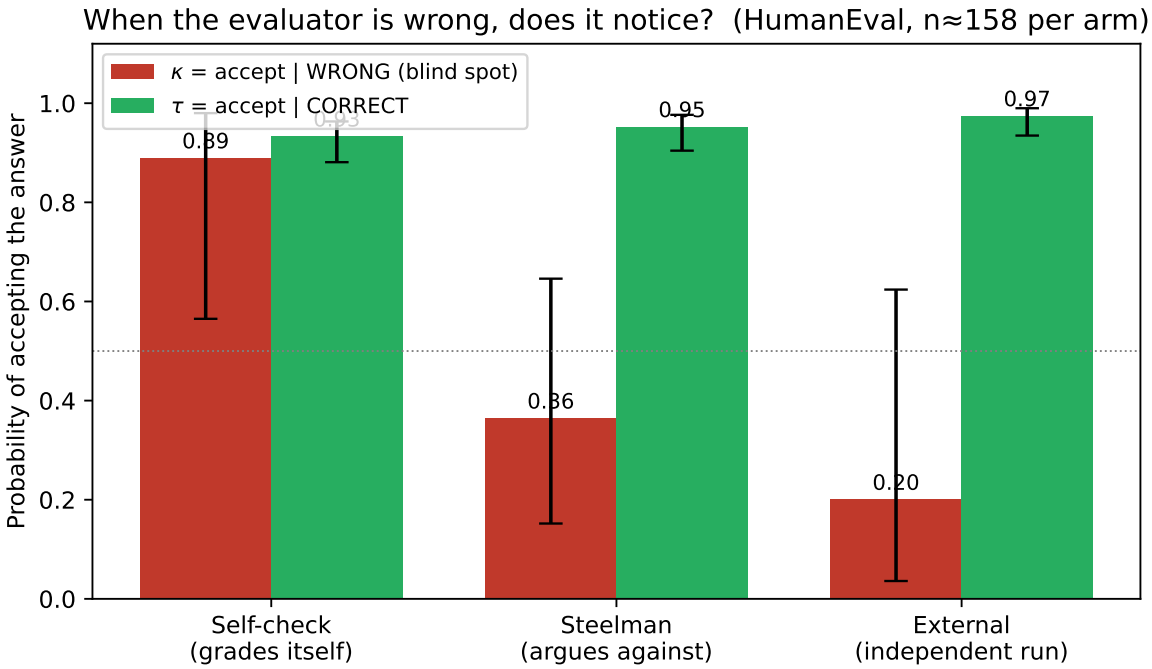
HumanEval-164, one model (gpt-5.4), three ways of checking its own work

Drew — working notes

September 8, 2026

The one-sentence result

When the model checks its *own* wrong answer, it almost never catches it — it accepts its own mistakes $\kappa = 0.889$ (about 89% of the time). Force it to argue *against* itself (steelman) and that drops to 0.364. Run a genuinely *independent* check and it drops to 0.200. The blind spot the paper predicts is real, and independence is what fixes it.



The numbers

arm	n	κ (accept|wrong)	τ (accept|correct)	$\Delta = \tau - \kappa$	$\Lambda = \tau/\kappa$	$E = \log_2 \Lambda$
Self-check	158	0.889 [0.565, 0.980]	0.933	0.044	1.05	0.07 bits
Steelman	157	0.364 [0.152, 0.646]	0.952	0.588	2.62	1.39 bits
External	157	0.200 [0.036, 0.624]	0.974	0.774	4.87	2.28 bits

Brackets are 95% confidence intervals (explained below). κ is the quantity the paper cares about: the probability the evaluator accepts an answer that is actually **wrong**. You want κ *low*. τ is the probability it accepts an answer that is actually correct; you want τ *high*. A good evaluator has a big gap Δ between them.

What a confidence interval (CI) actually is — the honest version

You ran the test *once* and got one number, e.g. $\kappa = 0.889$. But that number came from a *sample*. If you re-ran the whole experiment on a fresh batch of problems, you'd get a slightly different κ. The 95% CI is the range that, *if you repeated this experiment many times, would contain the true value about 95% of the time.*

The intuition that matters for you:

- **Narrow CI = you measured it well. Wide CI = you barely measured it.** The width is driven almost entirely by sample size n.
- CI width shrinks like $1/\sqrt{n}$. To make an interval *half* as wide you need *four times* the data, not twice. That is why the last bit of precision is always expensive.
- Here the κ intervals are wide — [0.57, 0.98], [0.15, 0.65], [0.04, 0.62] — because the model only got 9, 11, and 5 problems wrong in the three arms. You cannot estimate "how often does it catch its own errors" precisely when there are only a handful of errors to catch. **This is the main limitation and it belongs in the paper's Limitations section, stated plainly.**

Why "Wilson" intervals specifically

The textbook CI for a proportion is $\hat{p} \pm 1.96\sqrt{\hat{p}(1-\hat{p})/n}$ (the "normal" or Wald interval). It breaks badly when $\hat{p}$ is near 0 or 1, or n is small — exactly our situation (κ near 0.9 and 0.2, n of 5–11). The Wald interval can even produce impossible endpoints below 0 or above 1. The **Wilson score interval** is a small correction that stays valid for small n and extreme $\hat{p}$, and never leaves $[0, 1]$. It is the right default for paired binary data at this sample size, and it costs nothing — same formula complexity, just better behaved.

Does the difference actually mean anything? (McNemar)

The bars above compare *separate* arms. A sharper question: on the *same* problems, does switching the checker flip wrong→right more often than right→wrong? That is **McNemar's test**, and it uses only the "discordant" problems where the two arms disagree.

comparison	wrong→right	right→wrong	exact p
Self-check → Steelman	11	5	0.210
Self-check → External	13	1	**0.0018**
Steelman → External	10	4	0.180

Read it like this: moving from self-check to an external check, the external checker *fixed* 13 problems that self-check had wrongly accepted, and only *broke* 1 that self-check had right. 13-vs-1 is lopsided enough that it is very unlikely to be luck ($p = 0.0018$). The other two comparisons are closer to even (11-vs-5, 10-vs-4) and are *not* statistically significant — we cannot confidently say steelman beats plain self-check on final correctness at this sample size.

What p means, in one line

p = "if there were really *no* difference between the two checkers, how likely is a split this lopsided (or more) by pure chance?" Small p = the result would be a surprising coincidence under "no difference," so we take the difference seriously. It is *not* the probability that the result is real, and it says nothing about how *big* the effect is — only how surprising the pattern is under the null.

So what do we actually claim in the paper?

1. The self-evaluation blind spot (κ high) is real and large: $\approx$0.89 for a model grading its own work.
2. Independence reduces it substantially: κ 0.89 $\rightarrow$ 0.20, and the McNemar flip (13-vs-1, $p = 0.0018$) backs it.
3. We claim *operationalization and direction*, not proof of the theory, and we state the wide CIs and small error-count as the limitation. That is an honest, publishable result for JACIII.

Setup: HumanEval-164, model gpt-5.4, temperature 0, single-shot. $\kappa = P(\text{accept} \mid \text{wrong})$, $\tau = P(\text{accept} \mid \text{correct})$, $\Delta = \tau - \kappa$, $\Lambda = \tau/\kappa$, $E = \log_2 \Lambda$. *Arms: self-check (model grades itself), steelman (model argues against itself then grades), external (independent run grades). CIs are Wilson score intervals; paired comparison is exact McNemar.*